Python 3.15 Data Engineering Readiness Assessment

General Availability - Python 3.15.1 - tested 2026-10-26

94%

overall readiness | 15 PASS - 0 INCOMPAT - 2 BLOCKED - 0 SKIP - 0 FAIL (of 17 packages)

Executive summary

Python 3.15.1 was validated on macOS 27.0.1 ARM64 using uv. Of 17 packages probed, 15 passed and none failed outright. 2 did not pass: mlflow (BLOCKED); ray (BLOCKED).

Compared with 3.15.0 (2026-10-10), overall readiness moved from 92% to 94%.

Test environment

Release	3.15.1 (General Availability)
Python runtime	3.15.1
Python build	cpython-3.15.1-macos-arm64-none
Platform	macOS 27.0.1 ARM64
Test date	2026-10-26
Tester	Dr. Ceasar Jackson Jr.
Suite version	1.2.0
Docker image	pyarrow-dataeng:py314
Environment consistent	yes

Package results

Package	Status	Version	Notes
numpy	**PASS**	2.5.4	
pandas	**PASS**	3.1.0	
polars	**PASS**	1.41.2	
duckdb	**PASS**	2.0.0	
sqlalchemy	**PASS**	2.0.50	
pydantic	**PASS**	2.14.1	
matplotlib	**PASS**	3.11.2	
plotly	**PASS**	6.7.0	
jupyterlab	**PASS**	4.5.7	

Package	Status	Version	Notes
sqlite3	**PASS**	3.54.1	
pyarrow	**PASS**	26.0.0	
dask.dataframe	**PASS**	2026.7.1	
prefect	**PASS**	3.9.0	EventsWorker log errors reproduce identically on Python 3.14.10; not attributable to 3.15.1; root cause undetermined
mlflow	**BLOCKED**	3.17.2	Upstream pin pyarrow<26 plus no cp315 wheel for pyarrow 25.x. Workarounds: mlflow-skinny passes core tracking; full mlflow with a pyarrow>=26 override passes core, sklearn, dataset and table probes but fails uv pip check
ray	**BLOCKED**	unavailable	no cp315 wheels on the configured index
pyspark	**PASS**	4.2.0	Local mode only; PyArrow/pandas-backed features and cluster mode not tested
apache-airflow	**PASS**	3.3.2	dags test runs tasks in-process; scheduler and api-server not exercised. Binary-only resolution fails only because PyYAML 6.0.3 is sdist-only (source build passed 5/5 checks)

Changes since 3.15.0

Change	Package	Was	Now	Version
Improvement	prefect	INCOMPAT	**PASS**	3.9.0
Coverage change	apache-airflow	SKIP	**PASS**	3.3.2
Coverage change	mlflow	SKIP	**BLOCKED**	3.17.2

14 package(s) unchanged.

Coverage changes (moving into or out of SKIP) are reported separately from compatibility changes: a package that was not measured before is not evidence of a regression or an improvement in Python itself.

Evidence and limitations

- mlflow is BLOCKED: Upstream pin pyarrow<26 plus no cp315 wheel for pyarrow 25.x. Workarounds: mlflow-skinny passes core tracking; full mlflow with a pyarrow>=26 override passes core, sklearn, dataset and table probes but fails uv pip check.
- ray is BLOCKED: no cp315 wheels on the configured index.

How readiness is calculated

Readiness = (PASS + 0.50 x BLOCKED + 0.25 x INCOMPAT) / (packages - SKIP), rounded to a whole percent. SKIP is excluded because an unmeasured package is neither compatible nor incompatible. The weights are this project's convention, not an industry standard, and the score summarises this suite's probes only; it is not a certification of any package or deployment.